



USN

|  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|

**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
**Off-Campus Centre of Nitte (Deemed to be University)**  
**Seventh Semester B.Tech (CBCS) Degree Examinations**

December 2025

**EE3001-1 – INDUSTRIAL DRIVES & APPLICATIONS**  
(For EEE)

Duration: 3 Hours

Max. Marks: 100

**Note:**

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

**PART - A: MULTIPLE CHOICE QUESTIONS**

**20 Marks**

1. The usual major parts of an electrical drive include
  - A) Prime mover, gearbox, flywheel
  - B) Motor, power modulator, control unit, sensing unit, and source
  - C) Battery and alternator only
  - D) Compressor and turbine
2. A sensing unit in a drive is mainly used for
  - A) Increasing supply voltage
  - B) Cooling the motor
  - C) Feedback of variables like speed / current
  - D) Driver's safety
3. For traction applications, a desirable feature is
  - A) Poor low-speed torque
  - B) High torque at low speed and regenerative braking
  - C) No braking capability
  - D) Fixed-speed only
4. The steady operating point is where
  - A)  $T_e = T_L$  (including losses) at the given speed
  - B)  $T_e \gg T_L$
  - C)  $T_L \gg T_e$
  - D)  $dw/dt \neq 0$
5. Elevator/hoist load torque profile is strongly influenced by
  - A) Aerodynamic drag only
  - B) Counterweight, gravity, and inertial components
  - C) Only viscous friction
  - D) No external torque
6. The purpose of load equalization is to
  - A) Smooth out torque pulsations and reduce peak power demand
  - B) Increase average speed only
  - C) Increase torque ripple
  - D) Reduce inertia
7. Dynamic (rheostatic) braking in DC motor is achieved by
  - A) Reversing field polarity
  - B) Returning energy to grid
  - C) Connecting armature to a resistor with field excited
  - D) Running at no-load
8. The equivalent inertia seen by the motor in a belt drive depends on
  - A) Belt color
  - B) Square of speed ratio and reflected load inertia
  - C) Only load inertia
  - D) Ambient temperature
9. In a single-phase fully controlled rectifier feeding a DC motor, the average output voltage is controlled by
  - A) Armature resistance
  - B) Firing angle of thyristors
  - C) Supply frequency
  - D) Load torque

10. For a single-phase half-controlled rectifier, the conduction period per half cycle is
  - A)  $90^\circ$
  - B)  $(180^\circ - \alpha)$
  - C)  $120^\circ$
  - D) Depends on load
11. In a DC series motor fed from a rectifier, torque is proportional to
  - A) Armature current squared
  - B) Armature voltage
  - C) Field flux only
  - D) Speed squared
12. In multi-quadrant operation of DC drives, quadrant II corresponds to
  - A) Forward motoring
  - B) Forward braking
  - C) Reverse motoring
  - D) Reverse braking
13. A dual converter consists of
  - A) One bridge rectifier
  - B) Two fully controlled rectifiers in anti-parallel
  - C) Half-controlled rectifier
  - D) Diode rectifier
14. Current regulated voltage source inverter combines advantages of
  - A) VSI and CSI
  - B) VSI and rectifier
  - C) CSI and cycloconverter
  - D) VSI and DC motor
15. In Kramer drive, slip power is fed back to
  - A) Stator mechanically
  - B) Rotor mechanically
  - C) AC supply
  - D) DC motor
16. In closed loop control of induction motor drives, the feedback signal is generally
  - A) Voltage
  - B) Speed or torque
  - C) Frequency
  - D) Current only
17. An over-excited synchronous motor operates at
  - A) Lagging power factor
  - B) Leading power factor
  - C) Unity power factor
  - D) Variable power factor
18. Load commutated thyristor inverters are used for
  - A) Stepper motor control
  - B) DC series motor control
  - C) Self-controlled synchronous motor drives
  - D) Induction motor braking
19. Rolling mill drives require
  - A) High starting torque and precise speed control
  - B) Low torque
  - C) Constant voltage only
  - D) Stepper motor control
20. Which type of motor is preferred in rolling mills?
  - A) Slip-ring induction motor
  - B) Synchronous motor
  - C) Universal motor
  - D) Stepper motor

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

| Unit – I |                                                                                                                                                                            | Marks | BT* | CO* | PO* |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1.       | a) Draw the block diagram of an electrical drive system and explain its components.                                                                                        | 6     | L*2 | 1   | 1,2 |
|          | b) What are the reasons for using load equalization in an electric drive? Derive the equation for moment of inertia of flywheel used for load equalization.                | 6     | L3  | 1   | 1,2 |
|          | c) Half hour rating of a motor is 100kW. Heating time constant is 80 min and the maximum efficiency occurs at 70% full load. Determine the continuous rating of the motor. | 4     | L3  | 2   | 1,2 |
| 2.       | a) How to determine equivalent current and torque in case of fluctuating and intermittent load?                                                                            | 6     | L2  | 2   | 1,2 |
|          | b) Explain different components of load torques. Also give nature and classification of load torques.                                                                      | 4     | L2  | 1   | 1,2 |



- c) A constant speed drive has the following duty cycle:  
 Load rising from 0 to 400kW : 5 min  
 Uniform load of 500kW : 5 min  
 Regenerative power of 400kW returned to the supply: 4 min  
 Remains idle for: 2 min  
 Estimate power rating of the motor. Assume losses to be proportional to (power)<sup>2</sup>. 6 L3 2 1,2
3. a) Derive the equation for the overloading factor when the drive is subjected to short time duty 6 L2 2 1,2  
 b) Explain why regenerative braking fails in series motor. Demonstrate the dynamic braking applicable to dc separately excited and series motor. 4 L4 2 1,2  
 c) A drive has following parameters:  $J = 10 \text{ kg-m}^2$ ,  $T = 100 - 0.1N$ , N-m. Passive load torque  $T_l = 0.05N$ , N-m where  $N$  is the speed in rpm. Initially the drive is operating in steady- state. Now it is to be reversed. For this motor characteristics is changed to  $T = -100 - 0.1N$ , N-m. Calculate the time of reversal. 6 L3 1 1,2
- Unit – II**
4. a) With neat circuit diagram and necessary waveforms explain the working of single phase fully controlled rectifier control of dc separately excited motor. Derive the expression for output voltage. 8 L2 3 1,2  
 b) How different braking mechanism and multi-quadrant operation are realized in VSI Induction Motor drives? 8 L3 4 1,2
5. a) With neat circuit diagram explain the dual converter operation to obtain multi quadrant operation of dc separately excited motor. 8 L2 3 1,2  
 b) A Y-connected squirrel cage induction motor has the following ratings and parameters:  
 400V, 50Hz, 4 pole, 1370 rpm,  $R_s = 2 \Omega$ ,  $R_r' = 3 \Omega$ ,  $X_s = X_r' = 3.5 \Omega$ .  
 Motor is controlled by a voltage source inverter at constant v/f ratio. For regenerative braking operation of inverter fed induction motor drive determine  
 (i) Speed for the frequency of 30 Hz and 80% of full load torque  
 (ii) Frequency for a speed of 1000 rpm and full load torque.  
 (iii) Torque for a frequency of 40 Hz and speed of 1300 rpm. 8 L3 4 1,2
6. a) Explain how rotor resistance control can be used to control the induction motor drives. Explain static rotor resistance control using power electronic converter. 8 L3 4 1,2  
 b) A 230 V, 960 rpm, 200 A separately excited dc motor has an armature resistance of  $0.02 \Omega$ . It is fed from a chopper which provides both motoring and braking operations. The source has a voltage of 230 V. Assuming continuous conduction,  
 (i) Calculate duty ratio of chopper for motoring operation at rated torque and 350rpm.  
 (ii) Calculate duty ratio of chopper for braking operation at rated torque and 350rpm.  
 (iii) If maximum duty ratio of chopper is limited to 0.95 and maximum permissible motor current is twice the rated, calculate maximum permissible motor speed obtainable without field weakening and power fed to the source. 8 L3 3 1,2

## Unit – III

- |       |                                                                                                                                                                                                                                                                 |   |    |   |     |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|-----|
| 7. a) | Explain the following operation of synchronous motor drives subjected to fixed frequency supply,<br>(i) Starting (ii) Pull-in                                                                                                                                   | 7 | L3 | 5 | 1,2 |
| b)    | Explain the self-controlled synchronous motor drive employing load commutated thyristor inverter.                                                                                                                                                               | 7 | L3 | 5 | 1,2 |
| c)    | Explain the operational needs of textile mill drives and justify the choice of drive system used.                                                                                                                                                               | 2 | L4 | 5 | 1,2 |
| 8. a) | Explain the different stages and processes involved in a textile industry. Clearly indicate the types and ratings of the machines used in each section.                                                                                                         | 7 | L3 | 5 | 1,2 |
| b)    | Describe various processes involved in reversing cold rolling mill in a steel mill.                                                                                                                                                                             | 7 | L2 | 5 | 1,2 |
| c)    | Paper mills use several section drives that must operate synchronously to maintain paper tension. Analyze the consequences of improper speed synchronization between the press, drying, and winding sections and suggest how modern drives overcome this issue. | 2 | L4 | 5 | 1,2 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome.

\*\*\*\*\*